Unified Temporal Gravity (UTG): A Flat-Spacetime Field Theory with Measurement Structure

Abstract

We present a field-theoretic framework for gravitational phenomena defined on fixed Minkowski spacetime. The theory introduces a scalar field and a symmetric tensor field, with dynamics derived from a single action. Particle motion, signal propagation, and classical gravitational effects arise from the same variational principle. A measurement mapping from fields to detector observables yields a structured decomposition leading to four empirical relations. No spacetime curvature is introduced.

1 1. Background

We assume a fixed flat spacetime with metric: $\eta_{\mu\nu} = \mathrm{diag}(-1, 1, 1, 1)$

Coordinates: $\mathrm{x}^\mu = (t, x^i), \quad \mu = 0, 1, 2, 3, \quad i = 1, 2, 3$

Spacetime geometry is non-dynamical. No curvature tensors are used.

2 2. Fields

We define:

Scalar Field

$\chi(x^\mu) > 0$

Tensor Field

$\sigma_{\mu\nu}(x^\mu) = \sigma_{\nu\mu}$

with constraints: $\partial^\mu \sigma_{\mu\nu} = 0, \quad \sigma^\mu{}_\mu = 0$

All indices are raised/lowered using $\eta_{\mu\nu}$.

3 3. Action

The total action is: $\mathrm{S} = \int d^4x\, \mathcal{L}$
with Lagrangian density: $\mathrm{L} = 1\frac{}{2\partial_\mu\chi\partial^\mu\chi - V(\chi) + \frac{1}{2}\partial_\alpha\sigma_{\mu\nu}\partial^\alpha\sigma^{\mu\nu} + \lambda_1(\partial^\mu\sigma_{\mu\nu}) + \lambda_2(\sigma^\mu{}_\mu) + \chi\rho + \sigma_{\mu\nu}T^{\mu\nu}}$
Here:
* $\rho(x)$ is a scalar source * $T^{\mu\nu}(x)$ is a symmetric source tensor * λ_1, λ_2 enforce constraints

4 4. Field Equations

Applying Euler–Lagrange equations:

Scalar

$\partial_\mu\partial^\mu\chi = \frac{dV}{d\chi} - \rho$

Tensor

$\partial_\alpha\partial^\alpha\sigma_{\mu\nu} = T^{TT}_{\mu\nu}$
where TT denotes projection satisfying: $\partial^\mu\sigma_{\mu\nu} = 0, \quad \sigma^\mu{}_\mu = 0$

5 5. Particle Dynamics

A particle follows a worldline $x^\mu(\lambda)$.
Action: $\mathrm{S}_p = -m\int \chi(x)\sqrt{-\eta_{\mu\nu}\dot{x}^\mu\dot{x}^\nu}\, d\lambda$
Define: $^\mu = \frac{dx^\mu}{d\lambda}, \quad \dot{x}^2 = \eta_{\mu\nu}\dot{x}^\mu\dot{x}^\nu$
Variation yields: $\mathrm{d}\frac{}{d\lambda\left(\chi\frac{\dot{x}^\mu}{\sqrt{-\dot{x}^2}}\right) = \partial^\mu\chi\sqrt{-\dot{x}^2}}$

6 6. Tensor Coupling

We extend the action:
$\mathrm{S}_{\text{int}} = -m\int \chi(x)\left[1 + \alpha\frac{\dot{x}^\mu\dot{x}^\nu}{c^2(-\dot{x}^2)}\sigma_{\mu\nu}\right]\sqrt{-\dot{x}^2}\, d\lambda$
This yields the equation of motion: $\mathrm{d}^2x^\mu\frac{}{d\lambda^2 = -c^2\frac{\partial^\mu\chi}{\chi} - \alpha\partial^\mu\sigma_{\alpha\beta}\frac{\dot{x}^\alpha\dot{x}^\beta}{-\dot{x}^2}}$

7 7. Weak-Field Limit

Define: $\chi = 1 + \frac{\phi}{c^2}$
Then: $\nabla^2 \phi = \rho$
which reproduces Newtonian behavior.

8 8. Light Propagation

For massless motion: $^2 = 0$
Let $k^\mu = \frac{dx^\mu}{d\lambda}$.
Then: $\mathrm{d}^2 x^\mu \frac{}{d\lambda^2 = \partial^\mu \ln \chi + \alpha \partial^\mu \sigma_{\alpha\beta} k^\alpha k^\beta}$

9 9. Classical Effects

Light Deflection

$\Delta\theta = \frac{2GM}{bc^2}(1-\alpha)$
Matching observation: $\alpha = -1 \Rightarrow \Delta\theta = \frac{4GM}{bc^2}$

Perihelion Precession

$\Delta\theta = \frac{6\pi GM}{c^2 a(1-e^2)}$

Radiation

Far-field behavior: $\sigma_{ij} \sim \frac{1}{r}\ddot{Q}_{ij}$

$$P \propto \langle \dddot{Q}_{ij} \dddot{Q}_{ij} \rangle$$

10 10. Measurement Mapping

Detector output: $\mathrm{h}_a(t) = \mathcal{P}_a[\sigma_{\mu\nu}] + n_a(t)$
Two detectors: $\mathrm{H}_e(t), \quad L_e(t)$
Residual: $\mathrm{R}_e(t;\beta) = H_e(t) - \beta L_e(t)$

11 11. Signal Decomposition

Define: $\mathrm{S}_e = C_e + \lambda M_e$
$\mathrm{p}_C = \frac{C_e}{S_e}, \quad p_M = \frac{\lambda M_e}{S_e}$
$\mathrm{p}_C + p_M = 1$

12 12. State Representation

—$\Psi_e\rangle = \sqrt{p_C}|C\rangle + e^{i\Delta\phi}\sqrt{p_M}|M\rangle$

13 13. Interference

$\mathrm{I}_e = 2\sqrt{p_C p_M}\cos(\Delta\phi)$

14 14. Structural Consequences

From the above definitions:

$\mathrm{S}_e = F[R_e], \quad \Delta_e = G[R_e] \Rightarrow S_e \sim \Delta_e$

$\mathrm{p}_M \uparrow \Rightarrow |\Delta\phi| \uparrow$

—$\Delta\phi| \uparrow \Rightarrow S_e \downarrow$

$\mathrm{I}_e \sim \cos(\Delta\phi) \Rightarrow I_e \downarrow$ as $|\Delta\phi| \uparrow$

15 15. Conclusion

UTG provides a flat-spacetime field description of gravitational phenomena derived from a unified action. Classical results and structured measurement relations follow from the same framework without introducing curvature.